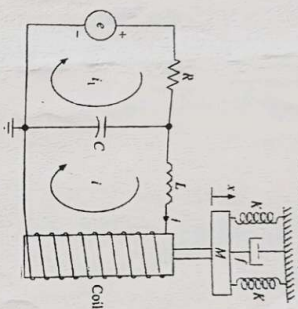


1.
 - a. What are the advantages and disadvantages of feedback in control systems? 2
 - b. Explain the construction and working of a synchro 3
2. Find the transfer function for the electro-mechanical system shown in Fig. below 3

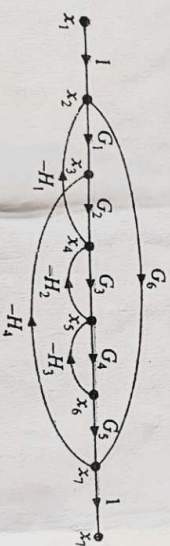


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3.
 - a. Compare ac and dc servomotor 2
 - b. A unity feedback system is characterized by an open loop transfer function 3

$G(s) = \frac{K}{s(s+10)}$. Determine the gain K so that the system has a damping ratio of 0.5. For this value of K , determine the settling time and peak overshoot.

4. Find the transfer function from the signal flow graph shown in the Fig. below



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